Maximus G1.5 Trouble Shooting And FAQ List

Contents

Maximus G1.5 Trouble Shooting And FAQ List 1

PMU2 - Overtemperature-Level1 6

PMU10 - PMU loop fault 6

PMU11 - PMU minor fault 6

PMU12 - PMU major fault 6

PMU13 - PMU no available battery 13

PMU35 - Overtemperature-Level2 14

PMU36 - Undervoltage 14

PMU37 - Overvoltage 14

PMU38 - Power supply output failure 14

PMU41 - No battery pack available 14

PMU46 - Relay（MOS） error 14

PMU47 - Pre-charge error 14

PMU48 - Pre-charge hardware error 15

PMU49 - Negative MOSFET temperature sensor error 15

PMU52 - Current sensor error 15

PMU57 - KSI Pre-MOS error 15

PMU59 - KSI MOS error 15

PMU61 - Battery pack 1 does not match 15

PMU62 - Battery pack 2 does not match 15

PMU63 - Battery pack 3 does not match 15

PMU64 - Battery pack 4 does not match 16

PMU65 - Battery pack 5 does not match 16

PMU66 - Battery pack 6 does not match 16

PMU67 - Abnormal charging status 16

TR2/TL2 - Drive motor stalled 16

TR4/TL4 - Motor overspeed protection 18

TR5/TL5 - Motor encoder error 18

TR6/TL6 - Controller phase loss 18

TR7/TL7 - MOSFET error 19

TR8/TL8 - Controller undervoltage 19

TR9/TL9 - Controller overvoltage 19
TR12/TL12 - Motor overtemperature 19
TR13/TL13 - Controller overtemperature 20
TR14/TL14 - Software Overcurrent 20
TR15/TL15 - Hardware Overcurrent 20
TR16/TL16 - Potentiometer error 21
TR19/TL19 - Operating sequence error 21
TR22/TL22 - Electromagnetic valve error 21
TR23/TL23 - Battery compartment (PMU)communication error 23
TR24/TL24 - Battery compartment (PMU) shutdown 24
TR25/TL25 - CAN timeout error - left drive 25
TR26/TL26 - CAN timeout error - right drive 26
TR27/TL27 - CAN timeout error - left blade 27
TR28/TL28 - CAN timeout error - middle blade 28
TR29/TL29 - CAN timeout error - right blade 29
TR31/TL31 - Seat switch verification error 29
TR32/TL32 - Software authentication error 30
TR60/TL60 - Controller undertemperature 30
TR61/TL61 - Power supply - 5V error 30
TR62/TL62 - Power supply - 12V error 30
TR63/TL63 - Motor temperature sensor error 31
TR64/TL64 - Controller flash error 31
TR65/TL65 - Motor encoder data error 31
TR66/TL66 - Controller internal communication error 32
TR67/TL67 - Motor over-temperature, power reduction 32
TR68/TL68 - Controller over-temperature, power reduction 32
TR69/TL69 - Left/Right parking brake manually disengaged and not reset 32
TR72/TL72 - Left/Right motor self-learning not completed 32
MR1/ML1/MM1 - Motor positioning error 32
MR2/ML2/MM2 - Drive motor stalled 33
MR3/ML3/MM3 - Blade motor low speed 33

MR4/ML4/MM4 - Blade motor over speed ... 34
MR6/ML6/MM6 - Controller phase loss ... 34
MR7/ML7/MM7 - MOSFET error ... 35
MR8/ML8/MM8 - Controller undervoltage ... 35
MR9/ML9/MM9 - Controller overvoltage ... 35
MR13/ML13/MM13 - Controller overtemperature ... 35
MR14/ML14/MM14 - Software overcurrent ... 36
MR15/ML15/MM15 - Hardware overcurrent ... 36
MR19/ML19/MM19 - Operating sequence error ... 37
MR25/ML25/MM25 - CAN timeout error - left drive ... 38
MR26/ML26/MM26 - CAN timeout error - right drive ... 39
MR27/ML27/MM27 - CAN timeout error - left blade ... 40
MR28/ML28/MM28 - CAN timeout error - middle blade ... 40
MR29/ML29/MM29 - CAN timeout error - right blade ... 41
MR31/ML31/MM31 - Blade switch verification error ... 41
MR32/ML32/MM32 - Software authentication error ... 42
D1 - CAN timeout with PMU ... 43
D2 - CAN timeout with left traction controller ... 44
D3 - CAN timeout with right traction controller ... 45
D4 - CAN timeout with left blade controller ... 46
D5 - CAN timeout with middle blade controller ... 47
D6 - CAN timeout with right blade controller ... 48

Tools： Computer、CAN DEBUG TOOL（ERP: R0203845-00）、16-Pin adapter（ERP: R0203075-00）、Jack、Multimeter.


CAN DEBUG TOOL（ERP: R0203845-00）| 16-Pin adapter（ERP: R0203075-00）


Jack | Multimeter

PMU2 - Overtemperature-Level1

Stop the vehicle for a period of time to allow it to cool down, if the fault still have after restarting, replace the battery compartment.

PMU10 - PMU loop fault

At least one channel in the battery compartment is unable to discharge, but the performance will not decrease, vehicle still can be worked.

If you feel a decrease in mowing performance with this fault code, the battery compartment should be replaced.

PMU11 - PMU minor fault

There have occurred a over-current(Level 1) or over-temperature(Level 1) fault. The vehicle's performance will be limited with this code, but this fault can be restored, no need to pay attention.

PMU12 - PMU major fault

Check：Use "ToolsForCAN" to identify the detail fault -

1. "PreCharge Function Error" of PMU12

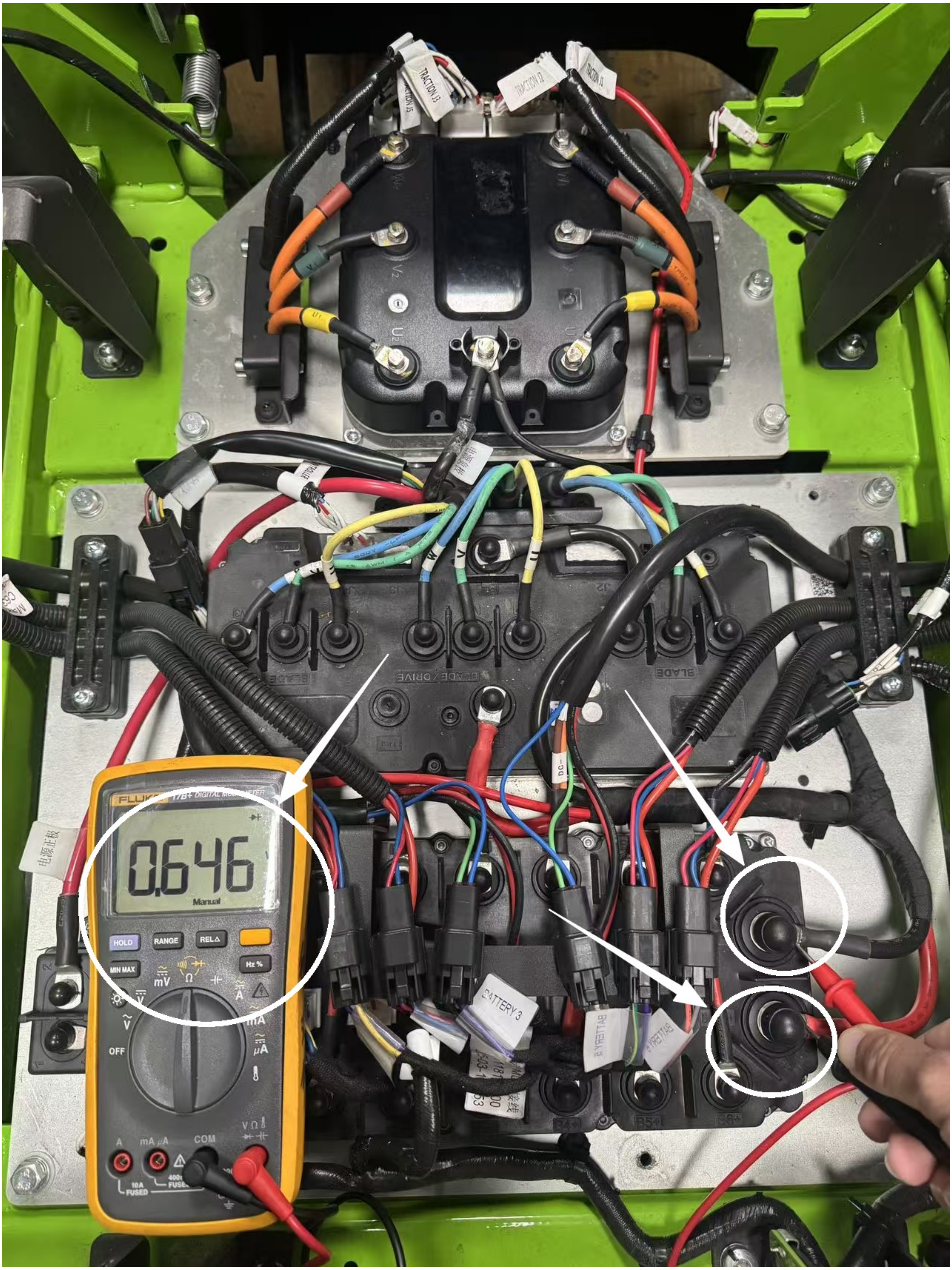
PMU FAULT LIST PART1

Function	Status	
Avg Curr Overcurr	◉ 0：Normal	○ 1：Avg Curr L1
	○ 2：Avg Curr L2	○ 3：Spare
PMUNeg. MOSOvertemp	◉ 0：Normal	○ 1：OvertempL1
	○ 2：OvertempL2	○ 3：NTCError
Peak Curr Overcurr	◉ 0：Normal	○ 1：Error
Feedback Curr Overcurr	◉ 0：Normal	○ 1：Error
PreCharge Function	○ 0：Normal	◉ 1：Error
PMUSoftware Auth	◉ 0：Normal	○ 1：Error
PMU Undervoltage Error	◉ 0：Normal	○ 1：Error
PMU Overvoltage Error	◉ 0：Normal	○ 1：Error

① Adjust the dial position of a multimeter to the diode position to measure the diode value of the PMU's B+ and B- as shown in the picture below(Red lead to black wire, black lead to red wire).

a. If the value is approximately equal to 0.7±0.2V， you should replace a new PMU.

b. Otherwise follow the steps.

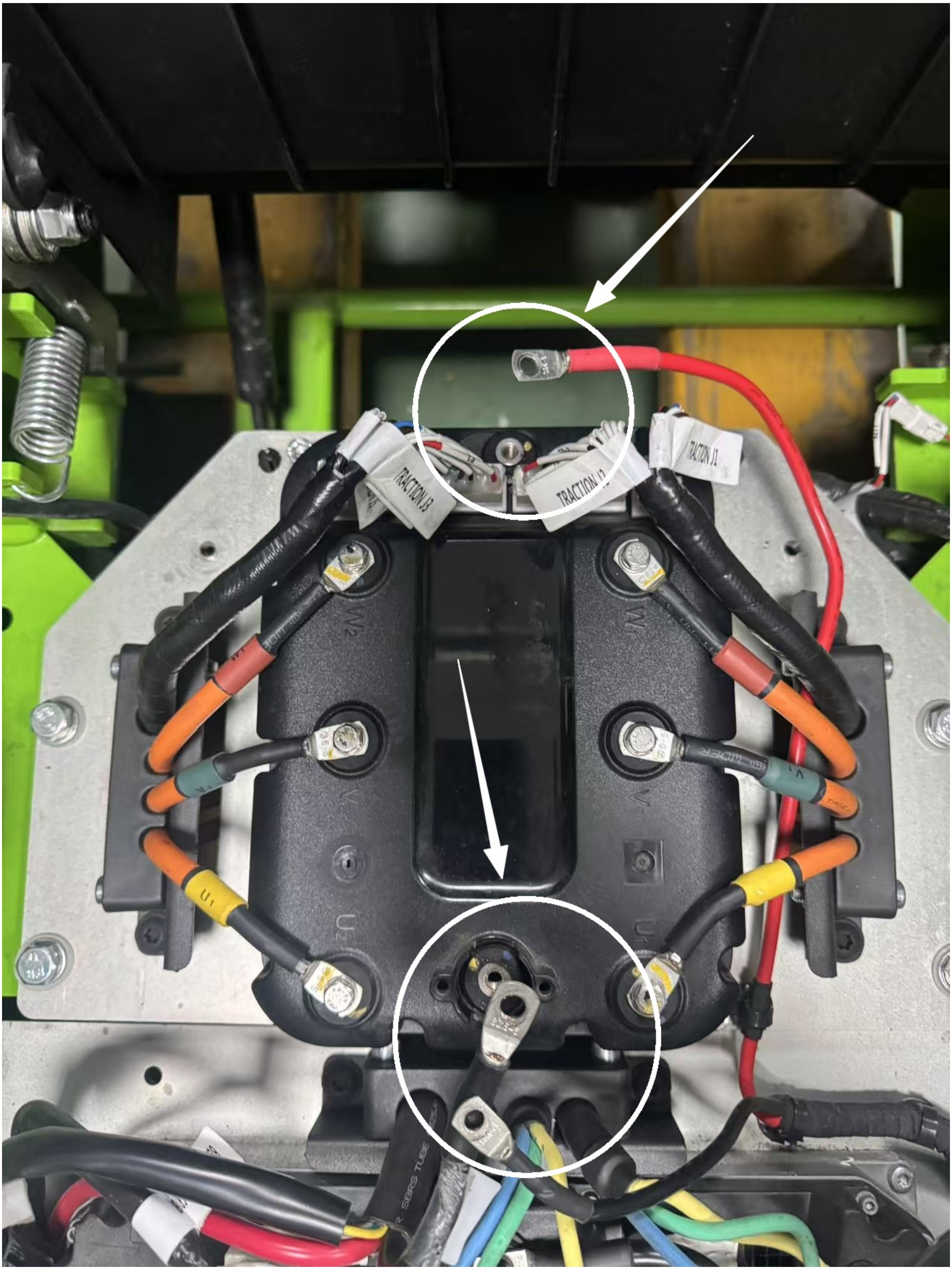

FLUKE
17B+ DIGITAL MULTIMETER
0.646
Manual
HOLD
RANGE
REL△
MIN MAX
Hz %
OFF
COM
BATTERY 3
BLADE/DRIVE
BLADE

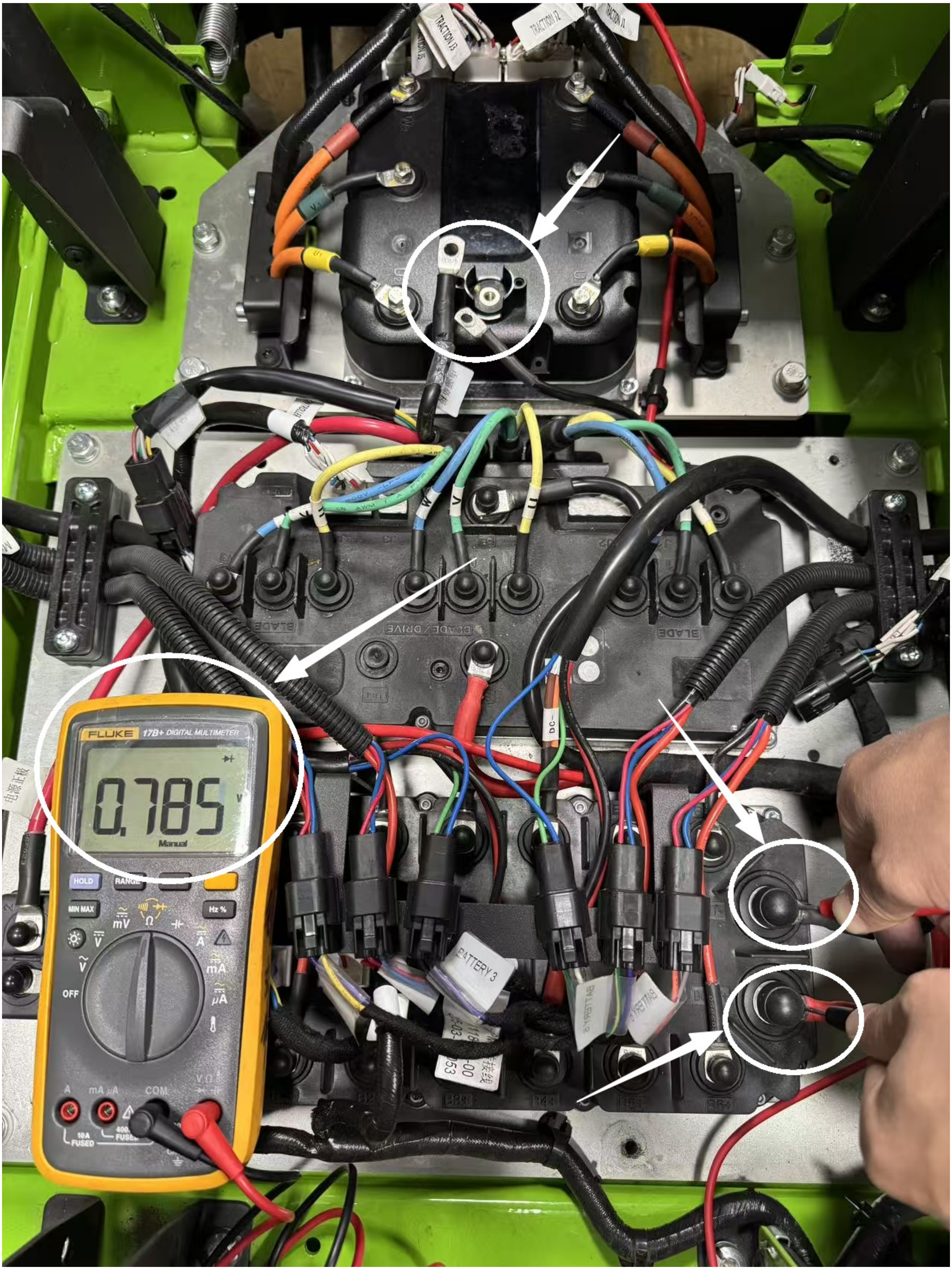
Correct value should be approximately equal to 0.7±0.2V

② Disconnect the traction controller's B+ and B- wires and use a multimeter to measure the diode value of the PMU's B+ and B- as shown in the picture below(Red lead to black wire, black lead to red wire).

a. If the value is approximately equal to 0.7±0.2V, you should replace the traction controller.

b. Otherwise, follow the steps.


TRACTION J3
TRACTION J2
TRACTION J1

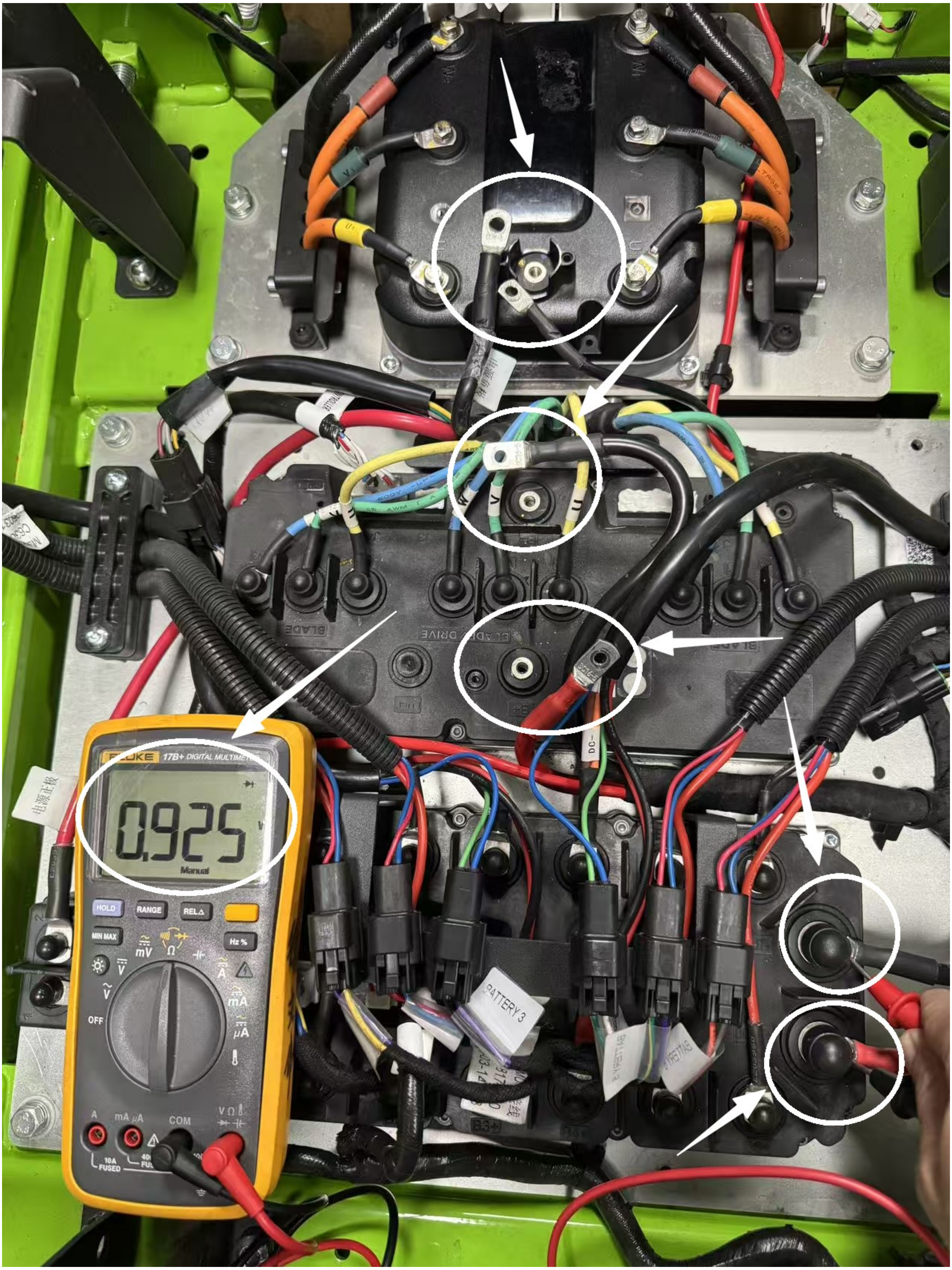

FLUKE 17B+ DIGITAL MULTIMETER
0.785
Manual
BATTERY 3
DC-
BLADE
BLADE/DRIVE

Correct value should be approximately equal to 0.7±0.2V

③ Disconnect the blade controller's B+ and B- wires and use a multimeter to measure the diode value of the PMU's B+ and B- as shown in the picture below(Red lead to black wire, black lead to red wire).

a. If the value is approximately equal to **0.7±0.3V**, you should replace the blade controller.

b. Otherwise, replace the PMU.


FLUKE 17B+ DIGITAL MULTIMETER
0.925
Manual
BATTERY 3

Correct value should be approximately equal to ***0.7±0.3V***

2. Other errors of PMU12

   Replace the PMU.

PMU13 - PMU no available battery

1. Make sure that the inserted batteries are correct, not mixed with different voltage platform.
2. Replace the battery compartment.

PMU35 - Overtemperature-Level2

Stop the vehicle for a period of time to allow it to cool down, if the fault still have after restarting, replace the battery compartment.

PMU36 - Undervoltage

1. Charge the batteries.
2. Make sure the battery compartment on the correct platform is used.
3. Replace the battery compartment.

PMU37 - Overvoltage

1. Make sure the battery compartment on the correct platform is used.
2. Replace the battery compartment.

PMU38 - Power supply output failure

Disconnect the power harness from battery compartment:

1. If the fault code disappeared, follow the “PMU12” trouble shooting guide to find the detail issue.
2. Otherwise, replace the battery compartment.

PMU41 - No battery pack available

1. Make sure that the inserted batteries are correct, not mixed with different voltage platform.
2. Replace the battery compartment.

PMU46 - Relay（MOS） error

Replace the battery compartment.

PMU47 - Pre-charge error

Follow the “PreCharge Function Error” of PMU12 trouble shooting guide to find the detail issue.

PMU48 - Pre-charge hardware error

Replace the battery compartment.

PMU49 - Negative MOSFET temperature sensor error

Replace the battery compartment.

PMU52 - Current sensor error

Replace the battery compartment.

PMU57 - KSI Pre-MOS error

Replace the battery compartment.

PMU59 - KSI MOS error

Replace the battery compartment.

PMU61 - Battery pack 1 does not match

1. Make sure that the inserted battery in cabin 1 is correct, not mixed with different voltage platform.
2. Replace the battery compartment.

PMU62 - Battery pack 2 does not match

1. Make sure that the inserted battery in cabin 2 is correct, not mixed with different voltage platform.
2. Replace the battery compartment.

PMU63 - Battery pack 3 does not match

1. Make sure that the inserted battery in cabin 3 is correct, not mixed with different voltage platform.
2. Replace the battery compartment.

PMU64 - Battery pack 4 does not match

1. Make sure that the inserted battery in cabin 4 is correct, not mixed with different voltage platform.
2. Replace the battery compartment.

PMU65 - Battery pack 5 does not match

1. Make sure that the inserted battery in cabin 5 is correct, not mixed with different voltage platform.
2. Replace the battery compartment.

PMU66 - Battery pack 6 does not match

1. Make sure that the inserted battery in cabin 6 is correct, not mixed with different voltage platform.
2. Replace the battery compartment.

PMU67 - Abnormal charging status

1. Check if the charger's input plug is properly plugged in.
2. Check if the vehicle can be powered on and worked properly:

   ① Can't be powered on and worked, then replace the battery compartment.

   ② Otherwise, replace the charge socket.

TR2/TL2 - Drive motor stalled

1. TR2

① Turn off the power, use a jack to lift the vehicle's rear wheels, disengage the manual parking brake, try to rotate the rear wheel of right side.

a. If the wheel can't be rotated, replace the right traction motor.

b. Otherwise follow the steps below.

② Swap the M5 terminals of left and right traction motors' phase harness(Totally six screws) and swap the 6 pins encoder sensor connectors(Traction J1 and Traction J5).

a. If still have TR2 fault code, replace the traction controller.

b. If fault code change to TL2, you should replace the right traction motor.

2. TL2

① Turn off the power, use a jack to lift the vehicle's rear wheels, disengage the manual parking brake, try to rotate the rear wheel of left side.

a. If the wheel can't be rotated, replace the left traction motor.

b. Otherwise follow the steps below.

② Swap the M5 terminals of left and right traction motors' phase harness(Totally six screws) and swap the 6 pins encoder sensor connectors(Traction J1 and Traction J5).

a. If still have TL2 fault code, replace the traction controller.

b. If fault code change to TR2, you should replace the left traction motor.

TR4/TL4 - Motor overspeed protection

1. TR4

Swap the M5 terminals of left and right traction motors' phase harness(Totally six screws) and swap the 6 pins encoder sensor connectors(Traction J1 and Traction J5).

a. If still have TR4 fault code, replace the traction controller.

b. If fault code change to TL4, you should replace the right traction motor.

2. TL4

Swap the M5 terminals of left and right traction motors' phase harness(Totally six screws) and swap the 6 pins encoder sensor connectors(Traction J1 and Traction J5).

a. If still have TL4 fault code, replace the traction controller.

b. If fault code change to TR4, you should replace the left traction motor.

TR5/TL5 - Motor encoder error

1. TR5

Swap the M5 terminals of left and right traction motors' phase harness(Totally six screws) and swap the 6 pins encoder sensor connectors(Traction J1 and Traction J5).

a. If still have TR5 fault code, replace the traction controller.

b. If fault code change to TL5, you should replace the right traction motor.

2. TL5

Swap the M5 terminals of left and right traction motors' phase harness(Totally six screws) and swap the 6 pins encoder sensor connectors(Traction J1 and Traction J5).

a. If still have TL5 fault code, replace the traction controller.

b. If fault code change to TR5, you should replace the left traction motor.

TR6/TL6 - Controller phase loss

1. TR6

Swap the M5 terminals of left and right traction motors' phase harness(Totally six screws) and swap the 6 pins encoder sensor connectors(Traction J1 and Traction J5).

a. If still have TR6 fault code, replace the traction controller.

b. If fault code change to TL6, you should replace the right traction motor.

2. TL6

Swap the M5 terminals of left and right traction motors' phase harness(Totally six screws) and swap the 6 pins encoder sensor connectors(Traction J1 and Traction J5).

a. If still have TL6 fault code, replace the traction controller.

b. If fault code change to TR6, you should replace the left traction motor.

TR7/TL7 - MOSFET error

Replace the traction controller.

TR8/TL8 - Controller undervoltage

① Charge the batteries.

② Make sure the controller on the correct platform is used.

③ Flash the right firmware to the traction controller. 60V and 80V platform have different firmware.

④ Replace the traction controller.

TR9/TL9 - Controller overvoltage

① Make sure the controller on the correct platform is used.

② Flash the right firmware to the traction controller. 60V and 80V platform have different firmware.

③ Replace the traction controller.

TR12/TL12 - Motor overtemperature

1. Stop the vehicle for a period of time to allow it to cool down, if the fault still have after restarting, follow the steps below.

Swap the M5 terminals of left and right traction motors' phase harness(Totally six screws) and swap the 6 pins encoder sensor connectors(Traction J1 and Traction J5).

① TR12

a. If still have TR12 fault code, replace the traction controller.

b. If fault code change to TL12, you should replace the left traction motor.

② TL12

a. If still have TL12 fault code, replace the traction controller.

b. If fault code change to TR12, you should replace the right traction motor.

TR13/TL13 - Controller overtemperature

Stop the vehicle for a period of time to allow it to cool down, if the fault still have after restarting, replace the traction controller.

TR14/TL14 - Software Overcurrent

1. TR14

Swap the M5 terminals of left and right traction motors' phase harness(Totally six screws) and swap the 6 pins encoder sensor connectors(Traction J1 and Traction J5).

a. If still have TR14 fault code, replace the traction controller.

b. If fault code change to TL14, you should replace the right traction motor.

2. TL14

Swap the M5 terminals of left and right traction motors' phase harness(Totally six screws) and swap the 6 pins encoder sensor connectors(Traction J1 and Traction J5).

a. If still have TL14 fault code, replace the traction controller.

b. If fault code change to TR14, you should replace the left traction motor.

TR15/TL15 - Hardware Overcurrent

1. TR15

Swap the M5 terminals of left and right traction motors' phase harness(Totally six screws) and swap the 6 pins encoder sensor connectors(Traction J1 and Traction J5).

a. If still have TR15 fault code, replace the traction controller.

b. If fault code change to TL15, you should replace the right traction motor.

2. TL15

Swap the M5 terminals of left and right traction motors' phase harness(Totally six screws) and swap the 6 pins encoder sensor connectors(Traction J1 and Traction J5).

a. If still have TL15 fault code, replace the traction controller.

b. If fault code change to TR15, you should replace the left traction motor.

TR16/TL16 - Potentiometer error

Check：Use "ToolsForCAN" to read the value of right/left potentiometer -

① If the value is larger than 4.6 or smaller than 0.2, replace the right/left potentiometer.

② Replace the traction controller.

③ Replace the control harness.

TR19/TL19 - Operating sequence error

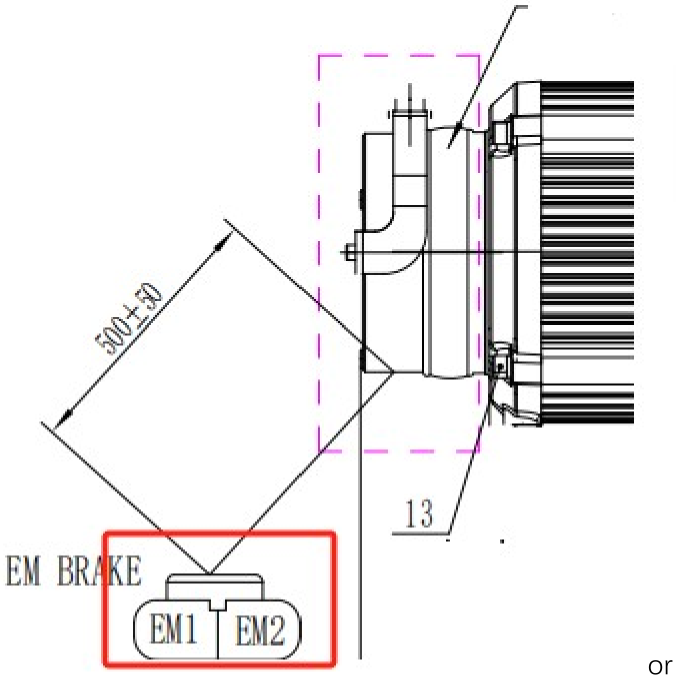
Check：Use "ToolsForCAN" to identify the status of switches and potentiometer -

Make sure there have no operations before sitting on the seat. The status should meet the table below before operating.

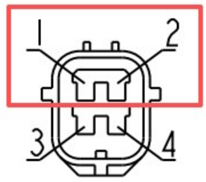
Switch & Controllers		State
Seat Switch		ON
Left Parking Switch		ON
Right Parking Switch		ON
PTO Switch		OFF
Left Potentionmeter	2. 40	2.3~2.5V
Right Potentionmeter	2. 40	2.3~2.5V
Left Traction Controller		No Fault
Right Traction Controller		No Fault
PMU		No Fault

TR22/TL22 - Electromagnetic valve error

Measure the resistance of the right/left electromagnetic brake.



or



刹车线/温度线接插件

名称	型号
刹车线外壳	174259-2
刹车线端子	171661-1
温保线端子	171631-1
刹车线密封圈	178210-1
刹车线锁片	174260-7

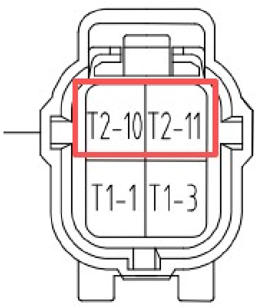
1. The value is within the range(40~200Ω).

Swap the left and right traction motors' EM brake connectors.

① If still have TR22/TL22 fault code

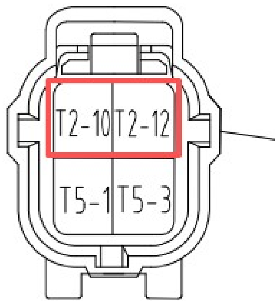
a. Check if the pins of the traction controller are bent or retracted, if can be fixed, try to fix them.

b. Check if the pins of the connector of the harness side are bent or retracted, if can be fixed, try to fix them, otherwise, replace the control harness.

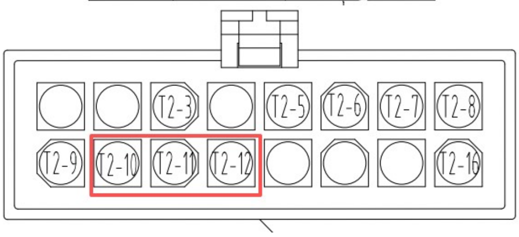


LEFT MOTOR EM BRAKE		
供应商	名称	型号
天海	外壳	0486801
	端子	0127102
	单线密封	2011404
	锁片	0486804

RIGHT MOTOR EM BRAKE		
供应商	名称	型号
天海	外壳	0486801
	端子	0127102
	单线密封	2011404
	锁片	0486804



TRACTION J2		
供应商	名称	型号
TE	外壳	794824-1
	端子	770904-1
力王	接口密封	1-1586362-6



c. Replace the traction controller.

② If TR22/TL22 disappeared and get a TL22/TR22 fault code, replace the right/left traction motor.

2. The value is not within the range(40~200Ω).

Replace the right/left traction motor.

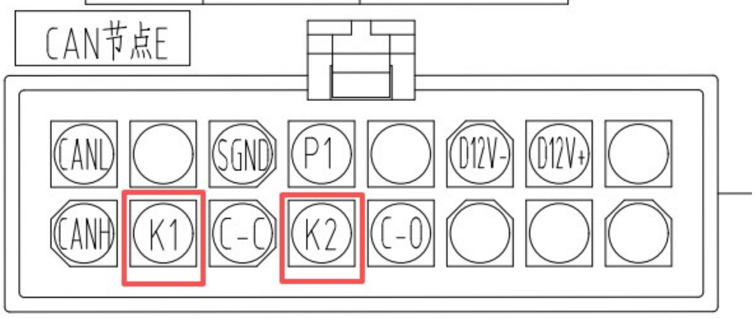
TR23/TL23 - Battery compartment (PMU)communication error

Check the pins of connector of the PMU side:

a. If the pins are bent or retracted, if can be fixed, try to fix them.

b. Replace the PMU.

PMU		
供应商	名称	型号
TE	外壳	794824-1
	端子	770904-1
力王	接口密封	1-1586362-6



TR24/TL24 - Battery compartment (PMU) shutdown

1. If you turn off the vehicle, you will get this fault code, it's normal.
2. Otherwise:

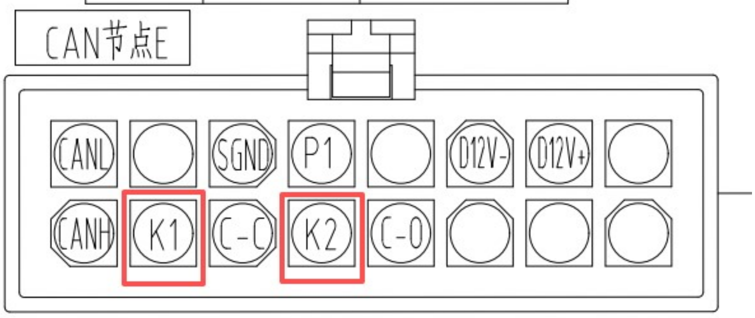
① Check if the connector of the key switch is connected well.

② Check the pins of connector of the PMU side:

a. If the pins of are bent or retracted, if can be fixed, try to fix them.

b. Replace the PMU.

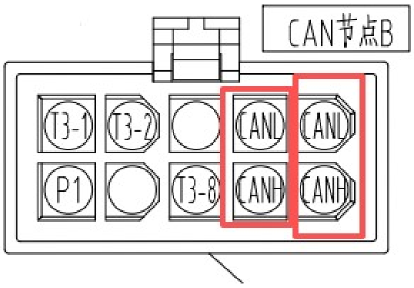
PMU		
供应商	名称	型号
TE	外壳	794824-1
	端子	770904-1
力王	接口密封	1-1586362-6



TR25/TL25 - CAN timeout error - left drive

Use multimeter to measure the resistance from the traction controller's 10 pins connector as the picture below.

TRACTION J3		
供应商	名称	型号
TE	外壳	794781-1
	端子	770904-1
力王	接口密封	1-794772-0



① If the resistance is around 60 Ohms(CANH to CANL):

a. Check if the pins of connector of controller side are bent or retracted, if can be fixed, try to fix them.

b. Replace the traction controller.

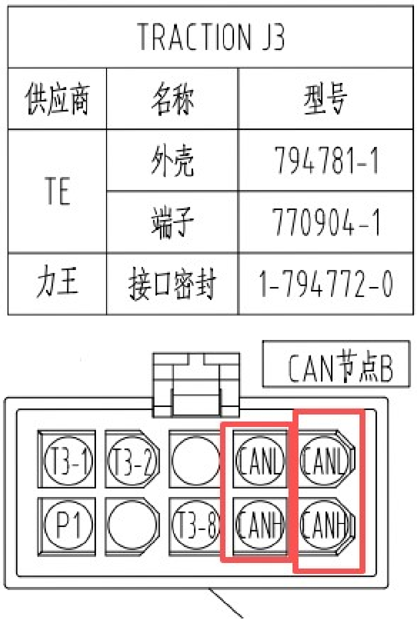
② Otherwise:

a. Check if the pins of the connector are bent or retracted, if can be fixed, try to fix them.

b. Replace the control harness.

TR26/TL26 - CAN timeout error - right drive

Use multimeter to measure the resistance from the traction controller's 10 pins connector as the picture below.



① If the resistance is around 60 Ohms(CANH to CANL):

a. Check if the pins of connector of controller side are bent or retracted, if can be fixed, try to fix them.

b. Replace the traction controller.

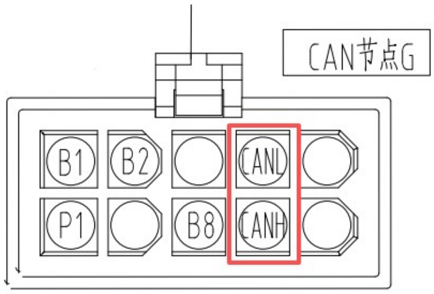
② Otherwise:

a. Check if the pins of the connector are bent or retracted, if can be fixed, try to fix them.

b. Replace the control harness.

TR27/TL27 - CAN timeout error - left blade

Use multimeter to measure the resistance from the blade controller's 10 pins connector as the picture below.



BLADE CONTROLLER		
供应商	名称	型号
TE	外壳	794781-1
	端子	770904-1
力王	接口密封	1-794772-0

① If the resistance is around 60 Ohms(CANH to CANL):

a. Check if the pins of connector of controller side are bent or retracted, if can be fixed, try to fix them.

b. Replace the blade controller.

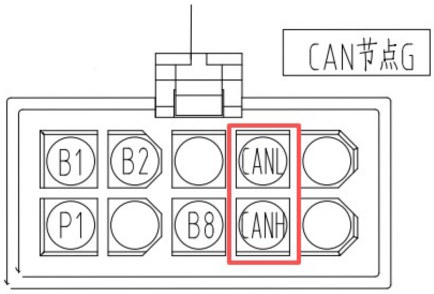
② Otherwise:

a. Check if the pins of the connector are bent or retracted, if can be fixed, try to fix them.

b. Replace the control harness.

TR28/TL28 - CAN timeout error - middle blade

Use multimeter to measure the resistance from the blade controller's 10 pins connector as the picture below.



BLADE CONTROLLER		
供应商	名称	型号
TE	外壳	794781-1
	端子	770904-1
力王	接口密封	1-794772-0

① If the resistance is around 60 Ohms(CANH to CANL):

a. Check if the pins of connector of controller side are bent or retracted, if can be fixed, try to fix them.

b. Replace the blade controller.

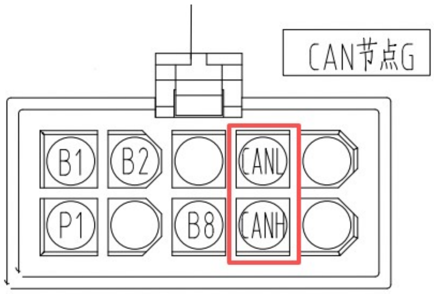
② Otherwise:

a. Check if the pins of the connector are bent or retracted, if can be fixed, try to fix them.

b. Replace the control harness.

TR29/TL29 - CAN timeout error - right blade

Use multimeter to measure the resistance from the blade controller's 10 pins connector as the picture below.



BLADE CONTROLLER		
供应商	名称	型号
TE	外壳	794781-1
	端子	770904-1
力王	接口密封	1-794772-0

① If the resistance is around 60 Ohms(CANH to CANL):

a. Check if the pins of connector of controller side are bent or retracted, if can be fixed, try to fix them.

b. Replace the blade controller.

② Otherwise:

a. Check if the pins of the connector are bent or retracted, if can be fixed, try to fix them.

b. Replace the control harness.

TR31/TL31 - Seat switch verification error

1. Check if the connector of the seat switch is connected well.
2. Check the pins of connector of the traction controller's side:

a. If the pins of the traction controller are bent or retracted, if can be fixed, try to fix them.

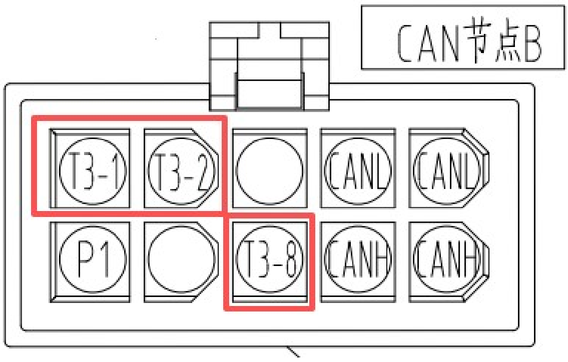
b. Replace the traction controller.

3. Check the pins of the connector of the harness side:

a. If the pins of are bent or retracted, if can be fixed, try to fix them.

b. Replace the control harness.

TRACTION J3		
供应商	名称	型号
TE	外壳	794781-1
	端子	770904-1
力王	接口密封	1-794772-0



4. Replace the seat switch.

TR32/TL32 - Software authentication error

Replace the traction controller.

TR60/TL60 - Controller undertemperature

1. Use the unit over -20℃/-4℉ .
2. Replace the traction controller.

TR61/TL61 - Power supply - 5V error

Replace the traction controller.

TR62/TL62 - Power supply - 12V error

Replace the traction controller.

TR63/TL63 - Motor temperature sensor error

This error will not cause a shutdown, but it will continue to display.

1. Check if the connectors of the traction motor are connected well.

2. TR63

Swap the 6 pins encoder sensor connectors(Traction J1 and Traction J5).

a. If still have TR63 fault code, replace the traction controller.

b. If fault code change to TL63, you should replace the right traction motor.

c. Otherwise, replace the control harness.

3. TL63

Swap the 6 pins encoder sensor connectors(Traction J1 and Traction J5).

a. If still have TL63 fault code, replace the traction controller.

b. If fault code change to TR63, you should replace the left traction motor.

c. Otherwise, replace the control harness.

TR64/TL64 - Controller flash error

Replace the traction controller.

TR65/TL65 - Motor encoder data error

1. Check if the connectors of the traction motor are connected well.

2. TR65

Swap the 6 pins encoder sensor connectors(Traction J1 and Traction J5).

a. If still have TR65 fault code, replace the traction controller.

b. If fault code change to TL65, you should replace the right traction motor.

c. Otherwise, replace the control harness.

3. TL65

Swap the 6 pins encoder sensor connectors(Traction J1 and Traction J5).

a. If still have TL65 fault code, replace the traction controller.

b. If fault code change to TR65, you should replace the left traction motor.

c. Otherwise, replace the control harness.

TR66/TL66 - Controller internal communication error

Replace the traction controller.

TR67/TL67 - Motor over-temperature, power reduction

The traction motor's temperature is too high, please reduce the load.

TR68/TL68 - Controller over-temperature, power reduction

The traction controller's temperature is too high, please reduce the load.

TR69/TL69 - Left/Right parking brake manually disengaged and not reset

Reengage the left/right parking brake.

TR72/TL72 - Left/Right motor self-learning not completed

1. Please finish the left/right motor self-learning.
2. Replace the left/right traction motor.
3. Replace the traction controller.

MR1/ML1/MM1 - Motor positioning error

1. MR1

Swap the M5 terminals of left and right blade motors' phase harness(Totally six screws).

a. If still have MR1 fault code, replace the blade controller.

b. If fault code change to ML1, you should replace the right blade motor.

2. ML1

Swap the M5 terminals of left and right blade motors' phase harness(Totally six screws).

a. If still have ML1 fault code, replace the blade controller.

b. If fault code change to MR1, you should replace the left blade motor.

3. MM1

Swap the M5 terminals of middle and right blade motors' phase harness(Totally six screws).

a. If still have MM1 fault code, replace the blade controller.

b. If fault code change to MR1, you should replace the middle blade motor.

MR2/ML2/MM2 - Drive motor stalled

1. MR2

① Turn off the power, try to rotate the right side blade.

a. If the blade can't be rotated or very tight, replace the right blade motor.

b. Otherwise follow the steps below.

② Swap the M5 terminals of left and right blade motors' phase harness(Totally six screws).

a. If still have MR2 fault code, replace the blade controller.

b. If fault code change to ML2, you should replace the right blade motor.

2. ML2

① Turn off the power, try to rotate the left side blade.

a. If the blade can't be rotated or very tight, replace the left blade motor.

b. Otherwise follow the steps below.

② Swap the M5 terminals of left and right blade motors' phase harness(Totally six screws).

a. If still have ML2 fault code, replace the blade controller.

b. If fault code change to MR2, you should replace the left blade motor.

3. MM2

① Turn off the power, try to rotate the middle side blade.

a. If the blade can't be rotated or very tight, replace the middle blade motor.

b. Otherwise follow the steps below.

② Swap the M5 terminals of middle and right blade motors' phase harness(Totally six screws).

a. If still have MM2 fault code, replace the blade controller.

b. If fault code change to MR2, you should replace the middle blade motor.

MR3/ML3/MM3 - Blade motor low speed

1. MR3

Swap the M5 terminals of left and right blade motors' phase harness(Totally six screws).

a. If still have MR3 fault code, replace the blade controller.

b. If fault code change to ML3, you should replace the right blade motor.

2. ML3

Swap the M5 terminals of left and right blade motors' phase harness(Totally six screws).

a. If still have ML3 fault code, replace the blade controller.

b. If fault code change to MR3, you should replace the left blade motor.

3. MM3

Swap the M5 terminals of middle and right blade motors' phase harness(Totally six screws).

a. If still have MM3 fault code, replace the blade controller.

b. If fault code change to MR3, you should replace the middle blade motor.

MR4/ML4/MM4 - Blade motor over speed

1. MR4

Swap the M5 terminals of left and right blade motors' phase harness(Totally six screws).

a. If still have MR4 fault code, replace the blade controller.

b. If fault code change to ML4, you should replace the right blade motor.

2. ML4

Swap the M5 terminals of left and right blade motors' phase harness(Totally six screws).

a. If still have ML4 fault code, replace the blade controller.

b. If fault code change to MR4, you should replace the left blade motor.

3. MM4

Swap the M5 terminals of middle and right blade motors' phase harness(Totally six screws).

a. If still have MM4 fault code, replace the blade controller.

b. If fault code change to MR4, you should replace the middle blade motor.

MR6/ML6/MM6 - Controller phase loss

1. MR6

Swap the M5 terminals of left and right blade motors' phase harness(Totally six screws).

a. If still have MR6 fault code, replace the blade controller.

b. If fault code change to ML6, you should replace the right blade motor.

2. ML6

Swap the M5 terminals of left and right blade motors' phase harness(Totally six screws).

a. If still have ML6 fault code, replace the blade controller.

b. If fault code change to MR6, you should replace the left blade motor.

3. ML6

Swap the M5 terminals of middle and right blade motors' phase harness(Totally six screws).

a. If still have MM6 fault code, replace the blade controller.

b. If fault code change to MR6, you should replace the middle blade motor.

MR7/ML7/MM7 - MOSFET error

Replace the blade controller.

MR8/ML8/MM8 - Controller undervoltage

1. Charge the batteries.
2. Make sure the controller on the correct platform is used.
3. Flash the right firmware to the blade controller. 60V and 80V platform have different firmware.
4. Replace the blade controller.

MR9/ML9/MM9 - Controller overvoltage

1. Make sure the controller on the correct platform is used.
2. Flash the right firmware to the blade controller. 60V and 80V platform have different firmware.
3. Replace the blade controller.

MR13/ML13/MM13 - Controller overtemperature

Stop the vehicle for a period of time to allow it to cool down, if the fault still have after restarting, replace the blade controller.

MR14/ML14/MM14 - Software overcurrent

1. MR14

Swap the M5 terminals of left and right blade motors' phase harness(Totally six screws).

a. If still have MR14 fault code, replace the blade controller.

b. If fault code change to ML14, you should replace the right blade motor.

2. ML14

Swap the M5 terminals of left and right blade motors' phase harness(Totally six screws).

a. If still have ML14 fault code, replace the blade controller.

b. If fault code change to MR14, you should replace the left blade motor.

3. MM14

Swap the M5 terminals of middle and right blade motors' phase harness(Totally six screws).

a. If still have MM14 fault code, replace the blade controller.

b. If fault code change to MR14, you should replace the middle blade motor.

MR15/ML15/MM15 - Hardware overcurrent

1. MR15

Swap the M5 terminals of left and right blade motors' phase harness(Totally six screws).

a. If still have MR15 fault code, replace the blade controller.

b. If fault code change to ML15, you should replace the right blade motor.

2. ML15

Swap the M5 terminals of left and right blade motors' phase harness(Totally six screws).

a. If still have ML15 fault code, replace the blade controller.

b. If fault code change to MR15, you should replace the left blade motor.

3. MM15

Swap the M5 terminals of middle and right blade motors' phase harness(Totally six screws).

c. If still have MM15 fault code, replace the blade controller.

d. If fault code change to MR15, you should replace the middle blade motor.

MR19/ML19/MM19 - Operating sequence error

1. Make sure the blade switch is on the “OFF” position, then try to turn on the blade again.
2. Check if the connector of the blade switch is connected well.
3. Check the pins of connector of the blade controller’s side:

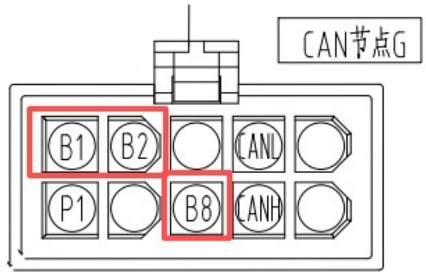
a. If the pins of the blade controller are bent or retracted, if can be fixed, try to fix them.

b. Replace the blade controller.

4. Check the pins of the connector of the harness side:

a. If the pins of are bent or retracted, if can be fixed, try to fix them.

b. Replace the control harness.



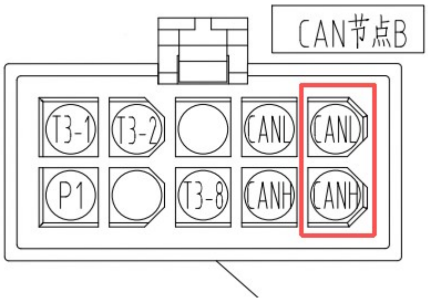
BLADE CONTROLLER		
供应商	名称	型号
TE	外壳	794781-1
	端子	770904-1
力王	接口密封	1-794772-0

5. Replace the blade switch,

MR25/ML25/MM25 - CAN timeout error - left drive

Use multimeter to measure the resistance from the traction controller's 10 pins connector as the picture below.

TRACTION J3		
供应商	名称	型号
TE	外壳	794781-1
	端子	770904-1
力王	接口密封	1-794772-0



① If the resistance is around 60 Ohms(CANH to CANL):

a. Check if the pins of connector of controller side are bent or retracted, if can be fixed, try to fix them.

b. Replace the traction controller.

② Otherwise:

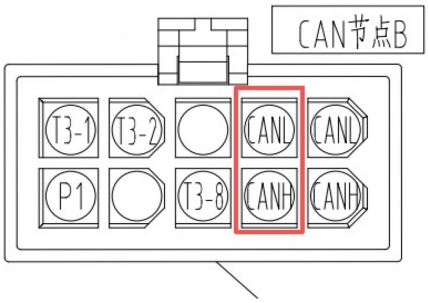
a. Check if the pins of the connector are bent or retracted, if can be fixed, try to fix them.

b. Replace the control harness.

MR26/ML26/MM26 - CAN timeout error - right drive

Use multimeter to measure the resistance from the traction controller's 10 pins connector as the picture below.

TRACTION J3		
供应商	名称	型号
TE	外壳	794781-1
	端子	770904-1
力王	接口密封	1-794772-0



① If the resistance is around 60 Ohms(CANH to CANL):

a. Check if the pins of connector of controller side are bent or retracted, if can be fixed, try to fix them.

b. Replace the traction controller.

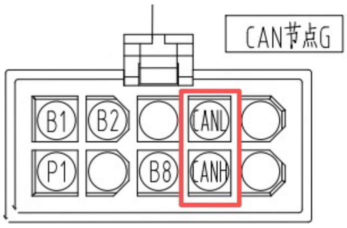
② Otherwise:

a. Check if the pins of the connector are bent or retracted, if can be fixed, try to fix them.

b. Replace the control harness.

MR27/ML27/MM27 - CAN timeout error - left blade

Use multimeter to measure the resistance from the blade controller's 10 pins connector as the picture below.



BLADE CONTROLLER		
供应商	名称	型号
TE	外壳	794781-1
	端子	770904-1
力王	接口密封	1-794772-0

① If the resistance is around 60 Ohms(CANH to CANL):

a. Check if the pins of connector of controller side are bent or retracted, if can be fixed, try to fix them.

b. Replace the blade controller.

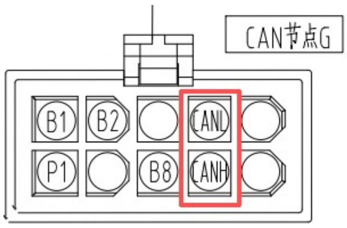
② Otherwise:

a. Check if the pins of the connector are bent or retracted, if can be fixed, try to fix them.

b. Replace the control harness.

MR28/ML28/MM28 - CAN timeout error - middle blade

Use multimeter to measure the resistance from the blade controller's 10 pins connector as the picture below.



BLADE CONTROLLER		
供应商	名称	型号
TE	外壳	794781-1
	端子	770904-1
力王	接口密封	1-794772-0

① If the resistance is around 60 Ohms(CANH to CANL):

a. Check if the pins of connector of controller side are bent or retracted, if can be fixed, try to fix them.

b. Replace the blade controller.

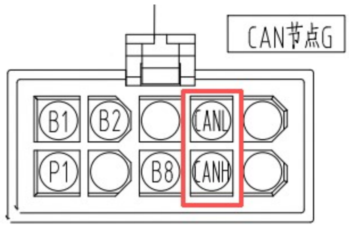
② Otherwise:

a. Check if the pins of the connector are bent or retracted, if can be fixed, try to fix them.

b. Replace the control harness.

MR29/ML29/MM29 - CAN timeout error - right blade

Use multimeter to measure the resistance from the blade controller's 10 pins connector as the picture below.



BLADE CONTROLLER		
供应商	名称	型号
TE	外壳	794781-1
	端子	770904-1
力王	接口密封	1-794772-0

① If the resistance is around 60 Ohms(CANH to CANL):

a. Check if the pins of connector of controller side are bent or retracted, if can be fixed, try to fix them.

b. Replace the blade controller.

② Otherwise:

a. Check if the pins of the connector are bent or retracted, if can be fixed, try to fix them.

b. Replace the control harness.

MR31/ML31/MM31 - Blade switch verification error

1. Check if the connector of the blade switch is connected well.
2. Check the pins of connector of the blade controller's side:

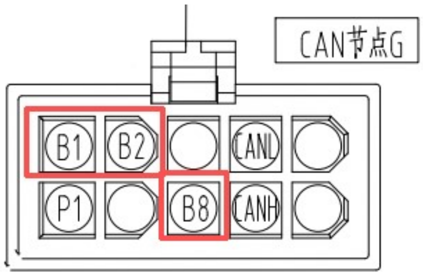
a. If the pins of the blade controller are bent or retracted, if can be fixed, try to fix them.

b. Replace the blade controller.

3. Check the pins of the connector of the harness side:

a. If the pins of are bent or retracted, if can be fixed, try to fix them.

b. Replace the control harness.



BLADE CONTROLLER		
供应商	名称	型号
TE	外壳	794781-1
	端子	770904-1
力王	接口密封	1-794772-0

4. Replace the blade switch,

MR32/ML32/MM32 - Software authentication error

Replace the blade controller.

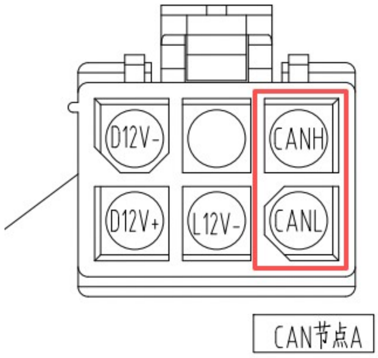
D1 - CAN timeout with PMU

1. If this fault exists simultaneously with any of D2/D3/D4/D5/D6, follow the steps below.

a. Disconnect the connector of display, use a multimeter to measure the resistance from “CANH” and “CANL”, if the value is approximately equal to 60Ω, replace the display.

b. Otherwise, the “CANH” and “CANL” of the control harness should be damaged, please replace a new control harness.

DISPLAY		
供应商	名称	型号
TE	外壳	172168-1
	端子	770988-1

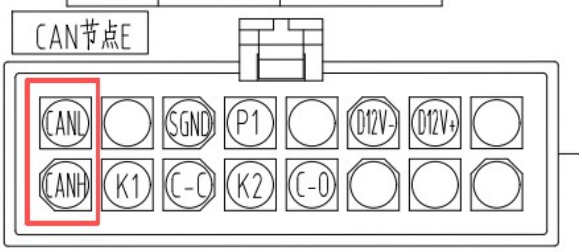


2. If this fault exists alone without D2/D3/D4/D5/D6, follow the steps below.

a. Disconnect the connector of PMU side, use a multimeter to measure the resistance from “CANH” and “CANL”, if the value is approximately equal to 60Ω, replace the PMU.

b. Otherwise, the “CANH” and “CANL” of the control harness should be damaged, please replace a new control harness.

PMU		
供应商	名称	型号
TE	外壳	794824-1
	端子	770904-1
力王	接口密封	1-1586362-6



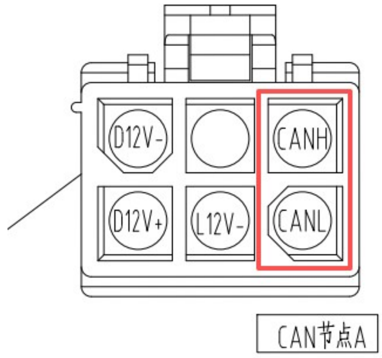
D2 - CAN timeout with left traction controller

1. If this fault exists simultaneously with D1, follow the steps below.

a. Disconnect the connector of display, use a multimeter to measure the resistance from “CANH” and “CANL”, if the value is approximately equal to 60Ω, replace the display.

b. Otherwise, the “CANH” and “CANL” of the control harness should be damaged, please replace a new control harness.

DISPLAY		
供应商	名称	型号
TE	外壳	172168-1
	端子	770988-1

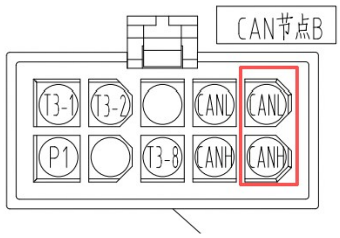


2. If this fault exists without D1, follow the steps below.

a. Disconnect the connector of traction controller side, use a multimeter to measure the resistance from “CANH” and “CANL”, if the value is approximately equal to 60Ω, replace the controller.

b. Otherwise, the “CANH” and “CANL” of the control harness should be damaged, please replace a new control harness.

TRACTION J3		
供应商	名称	型号
TE	外壳	794781-1
	端子	770904-1
力王	接口密封	1-794772-0



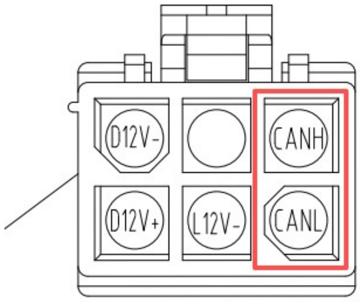
D3 - CAN timeout with right traction controller

1. If this fault exists simultaneously with D1, follow the steps below.

a. Disconnect the connector of display, use a multimeter to measure the resistance from “CANH” and “CANL”, if the value is approximately equal to 60Ω, replace the display.

b. Otherwise, the “CANH” and “CANL” of the control harness should be damaged, please replace a new control harness.

DISPLAY		
供应商	名称	型号
TE	外壳	172168-1
	端子	770988-1



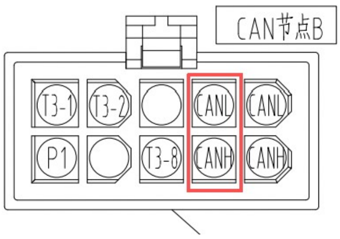
CAN节点A

2. If this fault exists without D1, follow the steps below.

a. Disconnect the connector of traction controller side, use a multimeter to measure the resistance from “CANH” and “CANL”, if the value is approximately equal to 60Ω, replace the controller.

b. Otherwise, the “CANH” and “CANL” of the control harness should be damaged, please replace a new control harness.

TRACTION J3		
供应商	名称	型号
TE	外壳	794781-1
	端子	770904-1
力王	接口密封	1-794772-0



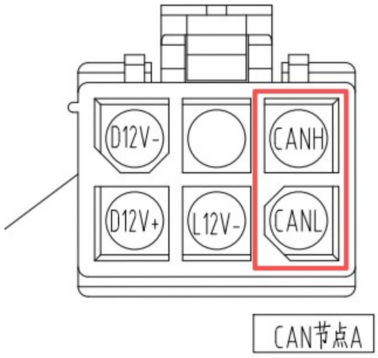
D4 - CAN timeout with left blade controller

1. If this fault exists simultaneously with D1, follow the steps below.

a. Disconnect the connector of display, use a multimeter to measure the resistance from “CANH” and “CANL”, if the value is approximately equal to 60Ω, replace the display.

b. Otherwise, the “CANH” and “CANL” of the control harness should be damaged, please replace a new control harness.

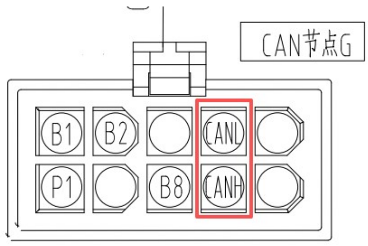
DISPLAY		
供应商	名称	型号
TE	外壳	172168-1
	端子	770988-1



1. If this fault exists without D1, follow the steps below.

a. Disconnect the connector of blade controller side, use a multimeter to measure the resistance from “CANH” and “CANL”, if the value is approximately equal to 60Ω, replace the controller.

b. Otherwise, the “CANH” and “CANL” of the control harness should be damaged, please replace a new control harness.



BLADE CONTROLLER		
供应商	名称	型号
TE	外壳	794781-1
	端子	770904-1
力王	接口密封	1-794772-0

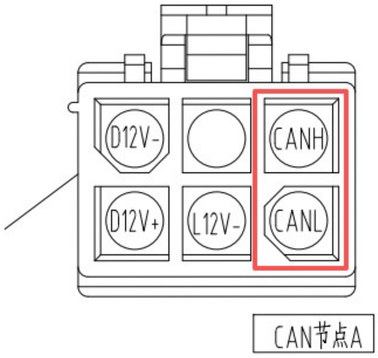
D5 - CAN timeout with middle blade controller

2. If this fault exists simultaneously with D1, follow the steps below.

c. Disconnect the connector of display, use a multimeter to measure the resistance from “CANH” and “CANL”, if the value is approximately equal to 60Ω, replace the display.

d. Otherwise, the “CANH” and “CANL” of the control harness should be damaged, please replace a new control harness.

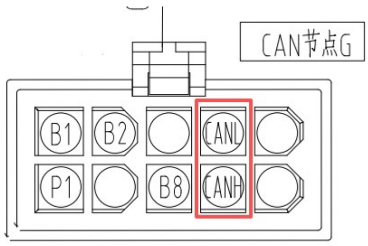
DISPLAY		
供应商	名称	型号
TE	外壳	172168-1
	端子	770988-1



2. If this fault exists without D1, follow the steps below.

c. Disconnect the connector of blade controller side, use a multimeter to measure the resistance from “CANH” and “CANL”, if the value is approximately equal to 60Ω, replace the controller.

d. Otherwise, the “CANH” and “CANL” of the control harness should be damaged, please replace a new control harness.



BLADE CONTROLLER		
供应商	名称	型号
TE	外壳	794781-1
	端子	770904-1
力王	接口密封	1-794772-0

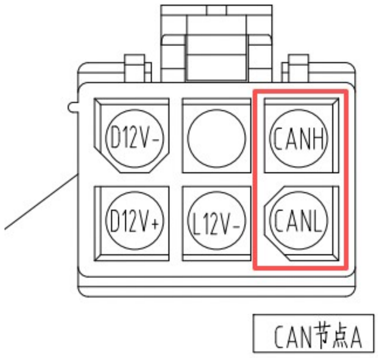
D6 - CAN timeout with right blade controller

1. If this fault exists simultaneously with D1, follow the steps below.

a. Disconnect the connector of display, use a multimeter to measure the resistance from “CANH” and “CANL”, if the value is approximately equal to 60Ω, replace the display.

b. Otherwise, the “CANH” and “CANL” of the control harness should be damaged, please replace a new control harness.

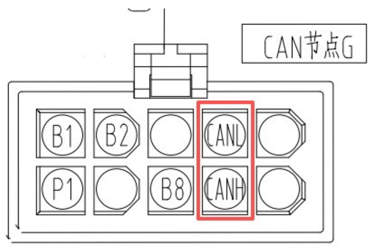
DISPLAY		
供应商	名称	型号
TE	外壳	172168-1
	端子	770988-1



2. If this fault exists without D1, follow the steps below.

a. Disconnect the connector of blade controller side, use a multimeter to measure the resistance from “CANH” and “CANL”, if the value is approximately equal to 60Ω, replace the controller.

b. Otherwise, the “CANH” and “CANL” of the control harness should be damaged, please replace a new control harness.



BLADE CONTROLLER		
供应商	名称	型号
TE	外壳	794781-1
	端子	770904-1
力王	接口密封	1-794772-0